

Assignment No. 2: Estimating Abalone Age

In this exercise, the Abalone dataset is used to predict the age of abalones. This dataset includes various features such as length, diameter, height, and different weight measurements of abalones, which are indirectly used to estimate their age. The age of abalones in this dataset is indicated by the "Rings" feature, which represents the age. Additionally, the gender feature, which is textual, will be ignored.

The goal of this exercise is to use the K-Nearest Neighbors (KNN) algorithm to predict the age of abalones. In this regard, you should test the KNN algorithm with different values of k (from 1 to 10) and examine its impact on model performance. To do this, first split the data into training and test sets, such that 70% of the data is allocated for training the model and 30% for evaluating model performance (test data).

After training the model with different k values, evaluate model accuracy using the test data. Use the Root Mean Squared Error (RMSE) metric to measure model accuracy. Finally, present the model accuracy for different k values in the form of a graph. Also, provide your analysis regarding model accuracy when dealing with different k values.

Dataset download link:

<https://archive.ics.uci.edu/dataset/1/abalone>